

Some Basic Concepts of Chemistry

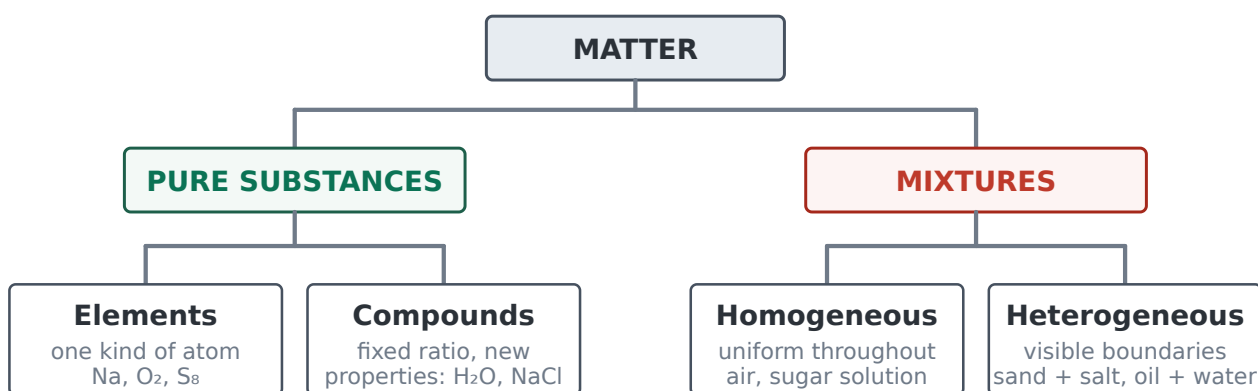
Complete chapter notes — every law, the full mole machinery, stoichiometry and solution concentration, with worked numericals and the traps that cost marks.

The spine of this chapter: chemistry counts particles but the laboratory can only weigh them. The **mole** is the bridge, and almost every numerical in this chapter — and in the whole two-year course — is the same three-step journey: **given quantity** → **moles** → **asked quantity**. Learn to spot which bridge (molar mass, Avogadro number, molar volume, molarity) each arrow needs and the chapter collapses to one idea.

1 • Matter, Properties and Measurement

1.1 Classification of matter

Matter is anything that has mass and occupies space. It is classified two ways — by **physical state** and by **chemical composition**.



Constituents of a mixture keep their own properties and can be separated by physical means;
constituents of a compound cannot — only a chemical reaction will break them apart.

Classification of matter by composition. The single most useful test in this diagram: can the parts be separated by a physical method (filtration, distillation, chromatography)? If yes it is a mixture; if it takes a chemical change, it is a compound.

1.2 Physical and chemical properties

Physical property	Chemical property
Measured without changing the identity of the substance	Observed only when the substance is converted into another substance
Colour, odour, melting point, boiling point, density, hardness	Combustibility, acidity, reactivity with water, rusting

1.3 SI units

Base quantity	Name	Symbol	Base quantity	Name	Symbol
Length	metre	m	Thermodynamic temperature	kelvin	K
Mass	kilogram	kg	Amount of substance	mole	mol
Time	second	s	Luminous intensity	candela	cd
Electric current	ampere	A			

Constants worth memorising: $1 \text{ L} = 1 \text{ dm}^3 = 10^3 \text{ cm}^3 = 10^{-3} \text{ m}^3$ · $1 \text{ bar} = 10^5 \text{ Pa}$ · $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ mm Hg}$ · density in SI is kg m^{-3} but in practice g cm^{-3} · $T(\text{K}) = t(^{\circ}\text{C}) + 273.15$ · $t(^{\circ}\text{F}) = \frac{9}{5} t(^{\circ}\text{C}) + 32$.

WHERE THIS LOSES MARKS

1. **Mass is not weight.** Mass is the quantity of matter (constant everywhere); weight is the force of gravity on it (varies with location).
2. The kelvin scale carries **no degree sign**: write 298 K, not 298°K. Negative kelvin temperatures do not exist.

2 · Handling Numbers

2.1 Precision and accuracy

Precision is how closely repeated measurements agree with *one another*; **accuracy** is how close a measurement is to the *true* value. A set of readings can be precise but inaccurate (a systematically mis-calibrated balance) — never assume one implies the other.

2.2 Significant figures

Significant figures are the digits known with certainty plus one final uncertain digit. They record how good a measurement is, so they must survive the arithmetic.

Rule	Example	Sig. figs
All non-zero digits are significant	285 cm	3
Zeros between non-zero digits are significant	2.005	4
Leading zeros are not significant — they only fix the decimal point	0.00234	3
Trailing zeros <i>after</i> a decimal point are significant	0.200	3
Trailing zeros in a whole number with <i>no</i> decimal point are ambiguous — use scientific notation	$100 \rightarrow 1.0 \times 10^2$	2
Exact numbers (counting, defined conversions) have infinite sig. figs	1 dozen = 12; 1 m = 100 cm	∞

Arithmetic rules. Addition/subtraction: the answer keeps the fewest *decimal places*.
Multiplication/division: the answer keeps the fewest *significant figures*.

MEMORY HOOK

Addition → **A**fter the point (decimal places). Multiplication → count the **M**eaningful digits (significant figures).

WORKED EXAMPLE 1 — SIGNIFICANT FIGURES IN ARITHMETIC

(a) $12.11 + 18.0 + 1.012 = 31.122 \Rightarrow \mathbf{31.1}$ (18.0 has only one decimal place).

(b) $2.5 \times 1.25 = 3.125 \Rightarrow \mathbf{3.1}$ (2.5 has only two significant figures).

Rounding off. If the digit to be dropped is greater than 5, raise the preceding digit by one; if less than 5, leave it; if it is exactly 5, raise the preceding digit only when it is **odd** (so 6.35 → 6.4, but 6.25 → 6.2).

2.3 Dimensional analysis (the factor-label method)

Multiply the given quantity by conversion factors written as fractions, arranged so that every unwanted unit cancels. The method never asks you to remember whether to multiply or divide — the units decide.

WORKED EXAMPLE 2 — UNIT CONVERSION

Convert 2.0 L to m^3 : $2.0 \text{ L} \times \frac{10^{-3} \text{ m}^3}{1 \text{ L}} = \mathbf{2.0 \times 10^{-3} \text{ m}^3}$.

Convert a density of 1.15 g cm^{-3} to kg m^{-3} : $\frac{1.15 \text{ g}}{\text{cm}^3} \times \frac{1 \text{ kg}}{10^3 \text{ g}} \times \frac{10^6 \text{ cm}^3}{1 \text{ m}^3} = \mathbf{1.15 \times 10^3 \text{ kg m}^{-3}}$.

3 · Laws of Chemical Combination

Five experimental laws were established before any atomic theory existed. Dalton's theory was invented afterwards, to explain them.

3.1 Law of conservation of mass (Lavoisier, 1789)

Matter can neither be created nor destroyed in a chemical reaction: **total mass of reactants = total mass of products**.

This is why chemical equations must be balanced, and it is the licence for every stoichiometric calculation you will ever do.

3.2 Law of definite (constant) proportions (Proust, 1799)

A given compound always contains exactly the **same elements in the same proportion by mass**, whatever its source or method of preparation.

Pure water from a river, from combustion of hydrogen, or from a laboratory synthesis is always 1 : 8 hydrogen to oxygen by mass. This law is what makes a chemical formula meaningful at all.

3.3 Law of multiple proportions (Dalton, 1803)

If two elements combine to form more than one compound, the masses of one element that combine with a **fixed mass** of the other are in a **simple whole-number ratio**.

WORKED EXAMPLE 3 — VERIFYING THE LAW OF MULTIPLE PROPORTIONS

Carbon and oxygen form CO and CO₂. Fix the carbon at 12 g:

In CO, 12 g C combines with **16 g** O. In CO₂, 12 g C combines with **32 g** O.

Ratio of the oxygen masses = 16 : 32 = **1 : 2**, a simple whole-number ratio. The law is obeyed.

Same treatment for H₂O and H₂O₂: fix 2 g of hydrogen; oxygen masses are 16 g and 32 g, again 1 : 2.

3.4 Gay-Lussac's law of gaseous volumes (1808)

When gases combine or are produced in a chemical reaction, they do so in a **simple whole-number ratio by volume**, provided all volumes are measured at the same temperature and pressure.

$\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ combines in the volume ratio 1 : 1 : 2 — exactly the ratio of the coefficients.

3.5 Avogadro's law (1811)

Equal volumes of all gases, at the same temperature and pressure, contain equal numbers of molecules.

Avogadro's insight was that the particles of an element need not be single atoms: taking hydrogen and oxygen to be *diatomic* resolved the contradiction between Gay-Lussac's volume ratios and Dalton's atomic theory. Its immediate consequence is that for gases, **volume ratio = mole ratio** — the reason gaseous stoichiometry can skip masses entirely.

FREQUENTLY ASKED

"Which law explains why gaseous reaction volumes are in simple ratios?" — Gay-Lussac's law is the *observation*; Avogadro's law is the *explanation*. Mention both, in that order.

4 · Dalton's Atomic Theory (1808)

1. Matter consists of indivisible atoms.
2. All atoms of a given element are identical in mass and properties; atoms of different elements differ.
3. Compounds form when atoms of different elements combine in fixed, simple whole-number ratios.
4. Chemical reactions only rearrange atoms — atoms are neither created nor destroyed.

What it explained: postulate 4 gives the law of conservation of mass; postulate 3 gives the laws of definite and multiple proportions.

LIMITATIONS YOU MUST BE ABLE TO STATE

1. It could not explain the law of gaseous volumes (that needed Avogadro's molecules).
2. Atoms are **not** indivisible — they contain electrons, protons and neutrons.
3. Atoms of the same element are **not** always identical in mass — **isotopes** exist (and isobars show different elements can share a mass).
4. It cannot explain why atoms bond at all, nor the existence of non-stoichiometric compounds.

5 · Atomic and Molecular Masses

5.1 The unified mass unit

1 u (unified mass unit, also called 1 dalton, Da) = $\frac{1}{12}$ of the mass of one atom of ^{12}C = 1.66056×10^{-24} g.

On this scale the ^{12}C atom is exactly 12 u *by definition*, hydrogen is 1.008 u and oxygen 16.00 u.

5.2 Average atomic mass

Because elements occur as mixtures of isotopes, the atomic mass in the periodic table is a **weighted average** over natural abundance:

$$\text{Average atomic mass} = \sum (\text{fractional abundance} \times \text{isotopic mass})$$

WORKED EXAMPLE 4 — AVERAGE ATOMIC MASS OF CHLORINE

Chlorine is 75.77% ^{35}Cl (34.9689 u) and 24.23% ^{37}Cl (36.9659 u).

$$\bar{A} = (0.7577)(34.9689) + (0.2423)(36.9659) = 26.4959 + 8.9568 = \mathbf{35.45 \text{ u}}$$

Notice the answer lies nearer 35 than 37 — the weighted average always leans towards the more abundant isotope. That single observation is enough to eliminate wrong options in an MCQ.

5.3 Molecular mass and formula mass

Molecular mass is the sum of the atomic masses of all atoms in a molecule: for CH_4 , $12.011 + 4(1.008) = 16.043$ u.

Formula mass is used for ionic compounds, which have no discrete molecules — NaCl is a continuous lattice, so we quote the formula mass $22.99 + 35.45 = 58.44$ u for the formula unit NaCl .

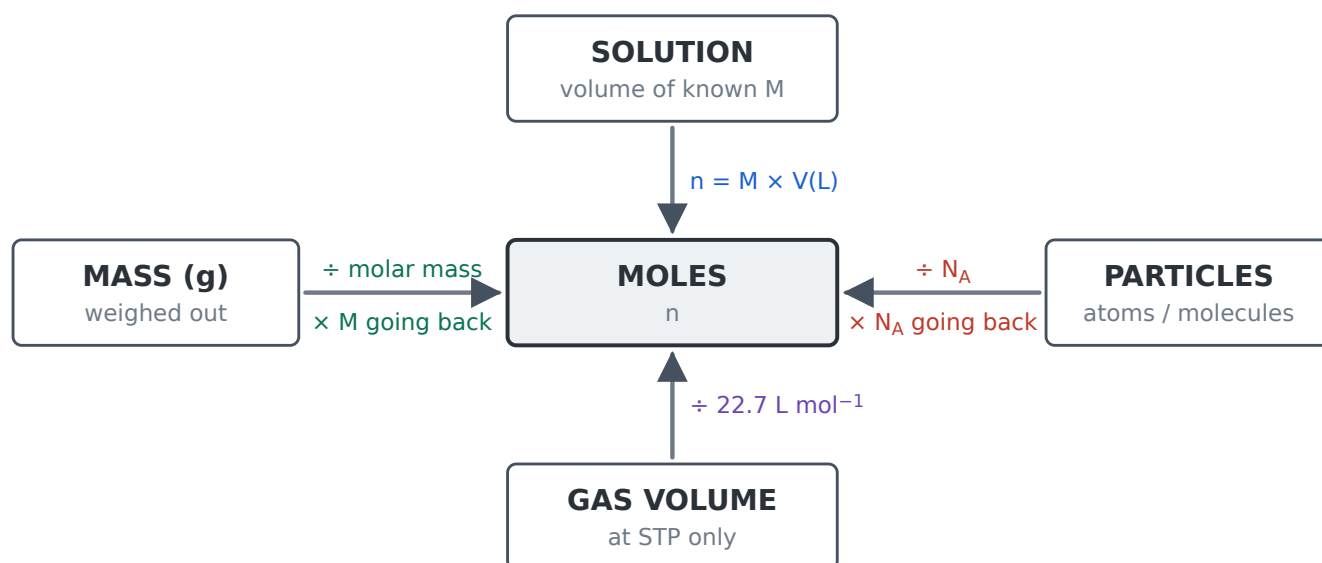
WHERE THIS LOSES MARKS

1. Writing "molecular mass of NaCl ". Ionic solids have a **formula mass**; there is no NaCl molecule.
2. Atomic/molecular mass is in **u** (a mass of one particle); molar mass is in **g mol^{-1}** (a mass of one mole). They are numerically equal, which is a deliberate design of the scale — not a coincidence you may ignore in the units.

6 · The Mole Concept

One **mole** is the amount of substance containing exactly 6.022×10^{23} elementary entities. $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ is the **Avogadro constant**.

The entity must always be named: one mole of O means 6.022×10^{23} oxygen *atoms*; one mole of O_2 means that many *molecules*, and therefore $2N_A$ atoms.



The mole map. Every numerical in this chapter is a path through this diagram. Reverse an arrow and the operation inverts: mass $\rightarrow$ mole divides by M , mole $\rightarrow$ mass multiplies by M .

6.1 Molar mass

$n = \frac{m}{M}$ where m is mass in g and M is molar mass in g mol^{-1} . Also $n = \frac{N}{N_A}$ and, for a gas at STP, $n = \frac{V}{22.7 \text{ L}}$.

6.2 Molar volume and the STP trap

22.7 L OR 22.4 L?

1. **STP is now 273.15 K and 1 bar**, at which the molar volume of an ideal gas is **22.7 L mol⁻¹** (22.71098 L). This is the value the current NCERT text uses.
2. The older **22.4 L mol⁻¹** applies at 273.15 K and **1 atm**. It is not wrong — it answers a different question.
3. Read the pressure stated in the question. If it says 1 bar use 22.7; if it says 1 atm use 22.4; state which you used.
4. Molar volume applies to **gases only**. There is no 22.7 L of liquid water.

WORKED EXAMPLE 5 — THE STANDARD MOLE CONVERSIONS

- (a) **Moles in 5.6 g of N₂:** $M = 28.0 \text{ g mol}^{-1}$, so $n = 5.6/28.0 = \mathbf{0.20 \text{ mol}}$.
- (b) **Molecules present:** $N = nN_A = 0.20 \times 6.022 \times 10^{23} = 1.2 \times 10^{23} \text{ molecules}$.
- (c) **Atoms present:** N₂ is diatomic, so $2 \times 1.2 \times 10^{23} = 2.4 \times 10^{23} \text{ atoms}$.
- (d) **Volume at STP:** $0.20 \times 22.7 = \mathbf{4.54 \text{ L}}$.

WORKED EXAMPLE 6 — COUNTING ATOMS INSIDE A COMPOUND

How many oxygen atoms are in 9.8 g of H₂SO₄?

$M(\text{H}_2\text{SO}_4) = 2(1.008) + 32.06 + 4(16.00) = 98.08 \approx 98 \text{ g mol}^{-1}$, so $n = 9.8/98 = 0.10 \text{ mol}$.

Each formula unit carries 4 oxygen atoms $\Rightarrow$ moles of O atoms = 0.40 mol.

$N = 0.40 \times 6.022 \times 10^{23} = 2.41 \times 10^{23} \text{ oxygen atoms}$.

7 · Percentage Composition and Formulae

7.1 Mass percentage of an element

$$\% \text{ element} = \frac{\text{total mass of that element in one mole of compound}}{\text{molar mass of the compound}} \times 100$$

For H₂O: $\% \text{H} = \frac{2 \times 1.008}{18.02} \times 100 = 11.19\%$ and $\% \text{O} = \frac{16.00}{18.02} \times 100 = 88.79\%$.

7.2 Empirical and molecular formula

Empirical formula	Molecular formula
The simplest whole-number ratio of atoms	The actual number of atoms in one molecule
CH for benzene; HO for hydrogen peroxide; CH ₂ O for glucose	C ₆ H ₆ ; H ₂ O ₂ ; C ₆ H ₁₂ O ₆

Molecular formula = (empirical formula) _{n} where $n = \frac{\text{molar mass}}{\text{empirical formula mass}}$. If $n = 1$ the two formulae coincide (H₂O, NH₃, CO₂).

METHOD 1 Percentage composition → molecular formula

1. Take **100 g** of the compound, so each percentage becomes a mass in grams.
2. Divide each mass by the atomic mass of that element to get **moles of atoms**.
3. Divide every result by the **smallest** of them — this gives the simplest ratio.
4. If any ratio is still fractional, multiply *all* of them by the smallest integer that clears it ($\times 2$ for .5, $\times 3$ for .33, $\times 4$ for .25). Write the empirical formula.
5. Compute the empirical formula mass, find $n = M/\text{EFM}$, and multiply every subscript by n .

The steps never change; only the arithmetic does. Step 4 is where most answers go wrong — never round 1.5 to 2.

WORKED EXAMPLE 7 — THE STANDARD NCERT NUMERICAL

A compound is 4.07% H, 24.27% C, 71.65% Cl by mass and has molar mass 98.96 g mol^{-1} . Find both formulae.

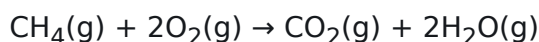
Element	Mass in 100 g	Moles	÷ smallest (2.021)
H	4.07 g	$4.07/1.008 = 4.04$	2.00
C	24.27 g	$24.27/12.01 = 2.021$	1.00
Cl	71.65 g	$71.65/35.45 = 2.021$	1.00

Empirical formula = **CH₂Cl**; empirical formula mass = $12.01 + 2(1.008) + 35.45 = 49.48$.

$n = 98.96/49.48 = 2 \Rightarrow$ molecular formula = **C₂H₄Cl₂**.

8 · Stoichiometry and the Limiting Reagent

A balanced equation is read as a **mole statement**, never a mass statement:



reads as "1 mole of methane reacts with 2 moles of dioxygen" — and, since all four are gases, also "1 volume with 2 volumes" by Avogadro's law. It does **not** read as 1 g with 2 g.

8.1 The limiting reagent

The **limiting reagent** is the reactant that is completely consumed first; it fixes the maximum amount of product. Every other reactant is in **excess** and is left over.

METHOD 2 Identifying the limiting reagent — the division test

1. Convert the given mass (or volume) of **each** reactant to moles.
2. Divide each reactant's moles by **its own stoichiometric coefficient** in the balanced equation.
3. The **smallest** quotient identifies the limiting reagent.
4. Base *all* product calculations on the limiting reagent alone.
5. Excess left over = (moles supplied) – (moles actually consumed, from the limiting reagent's ratio).

Never decide the limiting reagent by comparing raw masses, or even raw moles — only moles divided by coefficients are comparable.

WORKED EXAMPLE 8 — LIMITING REAGENT IN THE HABER PROCESS

50.0 kg of N_2 and 10.0 kg of H_2 are mixed: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. Find the mass of NH_3 formed.

$$n(\text{N}_2) = \frac{50.0 \times 10^3}{28.0} = 1786 \text{ mol} \rightarrow 1786/1 = 1786$$

$$n(\text{H}_2) = \frac{10.0 \times 10^3}{2.016} = 4960 \text{ mol} \rightarrow 4960/3 = 1653 \leftarrow \text{smaller, so H}_2 \text{ is limiting}$$

$$n(\text{NH}_3) = \frac{2}{3} \times 4960 = 3307 \text{ mol}; m = 3307 \times 17.03 = 5.63 \times 10^4 \text{ g} = \mathbf{56.3 \text{ kg of NH}_3}$$

N_2 consumed = $4960/3 = 1653 \text{ mol}$, so N_2 left over = $1786 - 1653 = 133 \text{ mol} = \mathbf{3.7 \text{ kg unreacted N}_2}$.

Note how badly the raw-mass instinct fails here: there is five times as much nitrogen by mass, and nitrogen is still the reactant in excess.

8.2 Percentage yield

$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$, where the theoretical yield is what the limiting reagent predicts.

9 · Concentration of Solutions

Throughout: the **solute** is the dissolved substance, the **solvent** is the medium, and mass of solution = mass of solute + mass of solvent.

Term	Definition	Formula	Depends on T ?
Mass percentage	Mass of solute per 100 g of solution	$\frac{w_B}{w_A + w_B} \times 100$	No
Mole fraction x_B	Moles of a component per total moles	$\frac{n_B}{n_A + n_B}$; $x_A + x_B = 1$	No
Molarity (M)	Moles of solute per litre of solution	$M = \frac{n_B}{V_{\text{solution}}(\text{L})}$	Yes
Molality (m)	Moles of solute per kilogram of solvent	$m = \frac{n_B}{w_A(\text{kg})}$	No
ppm	Parts of solute per 10^6 parts of solution	$\frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$	No

THE ONE-LINE REASON

Molarity is temperature-dependent because it is defined per unit **volume**, and volume expands on heating; molality is defined per unit **mass** of solvent, and mass does not change with temperature. Expect this as a one-mark question — and it is the reason molality is preferred in colligative-property work next year.

DERIVATION 1 Molarity from mass percentage and density

1. Take exactly **1 L = 1000 mL** of solution. Its mass is $1000 d$ g, where d is the density in g mL^{-1} .
2. If the solution is $x\%$ by mass, the mass of solute in it is $\frac{x}{100} \times 1000 d = 10 x d$ g.
3. Moles of solute = $\frac{10 x d}{M_B}$, where M_B is the molar mass of the solute.
4. This is contained in exactly 1 L, so it *is* the molarity.

$$M = \frac{10 \times (\text{mass \%}) \times d}{M_B} \text{ — with } d \text{ in } \text{g mL}^{-1}. \text{ The 10 comes from } 1000 \text{ mL} \div 100\%.$$

DERIVATION 2 Converting molarity to molality

1. Again take 1 L of solution: mass of solution = $1000 d$ g and moles of solute = M (numerically, by definition of molarity).
2. Mass of solute = $M \times M_B$ g.
3. Mass of **solvent** = mass of solution – mass of solute = $(1000 d - M M_B)$ g = $\frac{1000d - M M_B}{1000}$ kg.
4. Molality = moles of solute per kg of solvent = $\frac{M}{(1000d - M M_B)/1000}$.

$m = \frac{1000 M}{1000 d - M M_B}$. For dilute aqueous solutions $M M_B \ll 1000d$ and $d \approx 1$, so $m \approx M$ — which is why the two are often confused.

9.1 Dilution

Adding solvent changes the volume but not the moles of solute: $M_1 V_1 = M_2 V_2$. For mixing two solutions of the same solute: $M_{\text{final}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$.

WORKED EXAMPLE 9 — CONCENTRATED ACID TO MOLARITY

Commercial concentrated HCl is 38% by mass with density 1.19 g mL^{-1} . Find its molarity, and the volume needed to prepare 250 mL of 0.50 M HCl.

$$M = \frac{10 \times 38 \times 1.19}{36.46} = \mathbf{12.4 \text{ M}}$$

Dilution: $M_1 V_1 = M_2 V_2 \Rightarrow V_1 = \frac{0.50 \times 250}{12.4} = \mathbf{10.1 \text{ mL of the concentrated acid}}$, made up to 250 mL.

WORKED EXAMPLE 10 — MOLALITY AND MOLE FRACTION TOGETHER

2.5 g of ethanoic acid (CH_3COOH , $M = 60 \text{ g mol}^{-1}$) is dissolved in 75 g of benzene ($M = 78 \text{ g mol}^{-1}$). Find the molality and the mole fraction of the acid.

$$n_B = 2.5/60 = 0.0417 \text{ mol}; \quad n_A = 75/78 = 0.962 \text{ mol}.$$

$$m = \frac{0.0417}{0.075 \text{ kg}} = \mathbf{0.56 \text{ mol kg}^{-1}} \quad x_B = \frac{0.0417}{0.0417 + 0.962} = \mathbf{0.0416}$$

10 · Where marks are lost

THE TEN TRAPS OF CHAPTER 1

1. **Molar volume is 22.7 L at 1 bar, 22.4 L at 1 atm** — and applies to gases only. Say which standard you are using.
2. **Molarity is temperature-dependent; molality is not.** Molarity uses volume of *solution*, not of solvent.
3. **Never decide a limiting reagent from masses.** Moles $\div$ coefficient, then compare.
4. **Leading zeros are never significant; trailing zeros after a decimal point always are.** 0.0500 has three significant figures.
5. **Addition follows decimal places, multiplication follows significant figures.** Mixing the two rules is the commonest silent error.
6. **Do not round an empirical-formula ratio of 1.5 up to 2** — multiply every ratio by 2 instead.
7. **One mole of O_2 contains $2N_A$ atoms** but N_A molecules. Always ask what the question is counting.
8. **Average atomic mass is weighted by abundance**, never a plain mean of the isotopic masses.
9. **Ionic compounds have formula mass, not molecular mass.**
10. **Mass of solution = solute + solvent.** "75 g of benzene" is the solvent mass, so the solution weighs 77.5 g — a favourite one-mark deduction in molality problems.

Chapter 1 of the rationalised NCERT (2023-24 onward). Atomic masses used throughout: H 1.008, C 12.01, N 14.01, O 16.00, Cl 35.45, S 32.06, Na 22.99 u.